The story of the paper is that we can use universal categorical structures to give nice characterizations of metatheoretic proofs via abstract nonsense

The first approach to this uses curried displayed presheaves to characterize universal structures where the universal properties are stated with the path type

This runs into transport hell, which is also a huge performance issue

The solution is a hybrid approach that alleviates transport hell for these categorical semantics
↳ and ultimately enables gluing

The key design decisions are
- How we handle presheaves
- Use of inductive equality

Max and Eric B suggest that the story should instead be about compositional metatheory and orientation towards generality and future work
↳ and importantly compositionality

I should read Metatheory a la carte
↳ Delaware et al 2013

In this vein then, our contribution is a novel compositional approach to formalizing metatheoretic proofs

We use an analytic approach in the safe fragment of Cubical Agda, which is a reasonable general-purpose proof assistant

Mathematically, our framework fits into the following schematic:
- The universal property of syntax stated as a global section of displayed categories satisfying certain criteria
  ↳ Ex. The free Cartesian category CC[Σ] has a global section of any displayed category D over CC[Σ] with an interpretation $\iota : \Sigma \rightsquigarrow D$ that is displayed cartesian closed

$$\begin{tikzcd} D \arrow[d] \\ CC[\Sigma] \arrow[u, dashed, bend left, "elim_\iota"] \end{tikzcd}$$

CC^D $\iota : \Sigma \rightsquigarrow D$
(and $elim_\iota$ is a cartesian section)
not an equivalence?

Note that this is equivalent to the "usual" notion of freeness (as an initial object in the category of cartesian categories) by noting that displayed cartesianness is exactly the information guaranteeing that the total category of D is cartesian
↳ But the section presentation makes it so that the cartesian structure is definitionally over the free cartesian structure of CC[Σ]
↳ 1Lab makes a good case for this distinction on their normalization page

- We then can work modularly at the level of displayed universal properties

The "displayed" univprops are just univprops of the total category. We can go even further and describe vertical univprops modularly
↳ Vertical products/coproducts/terminal/initial/exponentials

- We make use of reindexing and prove theorems about preservation of displayed/vertical univprops along reindexing by certain functor
- Combining reindexing, preservation, and the global eliminator, we can define a local eliminator — the data necessary to provide a local section

$$\begin{tikzcd} & D^D \arrow[d] \\ CC[\Sigma] \arrow[r, "F"'] \arrow[ur, dashed, "localElim"] & D \end{tikzcd}$$

$\iota \rightsquigarrow D^D$
F cartesian
D^D vertically cartesian

This is the schematic for metatheoretic proofs (in this case for lang w/ prods)

The Secret Sauce

We have a generic characterization of universal properties as representability of certain presheaves.
↳ Stated as universal elements
↳ Why is this the best presentation?

Likewise, generic characterizations of displayed/vertical universal properties

These generic characterizations are the correct mathematical principles for conducting compositional gluing proofs but there is a slight issue: when instantiating this framework for a concrete category, we land in transport hell. This is not just a matter of incidental complexity and algebraic tedium of needing to manage transports. It is actually a performance issue
↳ Agda becomes unusably slow and frequently causes my machine to run out of memory
↳ Sometimes requesting up to 400 GB of memory

In the concrete cases we care about nearly all of the transports are over refl. In a theory w/ regularity, these would just go away. So to dispatch these transports, we provide a parallel characterization of the displayed/vertical universal properties in a system that is regular - namely, we replace the cubical path type with an inductive equality type
↳ By definition, J Eq.refl ≜ id

We then create coercions from the Eq univprops to the Path univprop. In this manner, the Eq characterization is a convenient interface to introduce concrete witnesses of these universal structures

The prereqs to use these Eq intro principles is that category laws (associativity and unitality) + naturality of each (π₁ and − × A) hold definitionally in the base category of the displayed category for which we are defining the universal strs

These properties readily hold in the category of sets, which suffices to prove canonicity (as thats the domain of interpretation for a canonicity proof)

For a richer metatheoretical proof, such as conservativity or normalization, we need to eliminate into a presheaf category

It is crucial to present presheaves favorably to obey the preconditions to our lemma
↳ By default it does not, but we can make it so by choosing an alternative representation of presheaves

Why universal elements are based

Look at binary products

Manual

```
record prod A B where
 field
   prod : ob
   π₁ : C[ob, A]
   π₂ : C[ob, B]
   univMap : C[X, A] → C[X, B]
     → C[X, ob]
   commutes₁ : f : C[x, A] → g : C[x, B]
     → univMap f g ⋆ π₁ ≡ f
   commutes₂ : "                "
   univMapUniq
```

Via representability

generic

the prod of A and B is an object P s.t. よP ≅ よA × よB

Via Universal element

← generic for Psh P

vert : Ob
elt : P-vert
isUniv : composition w/ elt is an equivalence

Equivalent to the representation presentation

Reduces the manual definition to a choice of presheaf, which is already latent in the manual definition anyway

Can state theorems/constructions about universal structures all at once/in general

Generalizes directly to the displayed setting and offers the same level of generality

Somehow the element captures the essence of the connective being defined
↳ i.e. products are negative types and are thus characterized by projections